25900518

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCMBG201 - MICROBIAL GENETICS

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Part A – Answer Any 10 10X2 = 20

1. **Which strain of Streptococcus pneumoniae caused the death of mice in Griffith's experiment? [An] / [CO1]**
   The **S strain** (Smooth strain**).[1.5] .** It was virulent due to the presence of a **polysaccharide capsule**. **[0.5]**

2. **Give the limitations of Griffith's experiment. [K]/[CO1]**
   It could not identify the **chemical nature** of the "transforming principle" (DNA) **[1].** It did not provide a **molecular mechanism** for how the transformation occurred **[1].**

3. **What are intercalating agents? Give one example [K]/[CO3]**
   Molecules that insert (intercalate) between the base pairs of a DNA double helix. **[1].** This insertion distorts the DNA structure, often leading to **frameshift mutations** during replication. **Ethidium bromide** is an example.**[1]**

4. **State one advantage of rolling circle replication. [K]/[CO1]**
   It allows for the rapid production of multiple copies of a circular DNA molecule (like a plasmid or viral genome) from a single origin.**[2]**

5. **Mention the steps involved in generalised transduction. [K] [CO2]**
   A bacteriophage infects a donor bacterial cell, initiating the lytic cycle. **Mistaken Packaging:** Bacterial DNA fragments are packaged into a phage capsid instead of viral DNA. **[1].** The transducing particle infects a new recipient bacterium. The bacterial DNA is injected into a new recipient cell and integrated into its genome**.[1]**

6. **Define Okazaki fragments. [K]/[CO1]**
   Okazaki fragments are short, newly synthesised DNA fragments formed on the lagging strand during DNA replication**. [2]**

7. **What is genetic recombination? [K] [CO2]**

   Genetic recombination is the process by which genetic material is broken and rejoined to produce new combinations of alleles or genes. **[2]**

8. **What are transposons? [K] [CO2]**

   DNA sequences that can change their position within a genome often duplicate themselves in the process.**[2]**

9. **Define excision repair in bacteria. [K] [CO3]**

   A DNA repair mechanism that removes a segment of a damaged DNA strand (e.g., containing a thymine dimer) and synthesises a new, correct strand using the complementary strand as a template. **[2]**

10. **What is the key difference between generalised and specialised transduction? [K] [CO2]**

    Generalised transduction transfers any random part of the bacterial chromosome. **[1]** Specialised transduction transfers only specific genes located near the prophage attachment site. **[1]**

11. **What are the steps of gene expression in bacteria [K] [CO2]**

    Transcription: The DNA sequence of a gene is copied into a complementary messenger RNA (mRNA) molecule. **[1].** Translation: The mRNA is decoded by a ribosome to synthesise a specific protein using tRNA molecules. **[1]**

12. **What is the purpose of the Ames test? [K] [CO3]**

    To identify potential **chemical mutagens [1]** by measuring their ability to cause reversions in histidine-requiring mutant strains of *Salmonella* bacteria. **[1]**

Part B – Short Essay - Answer Any 4. 4X5 = 20

13. **What are lethal mutations? How can conditional mutants help in studying lethal genes? [K] [CO3]**

    Lethal Mutation: A mutation that causes death, preventing study under normal conditions. **[2] Conditional Mutants -** Allow growth under **permissive conditions** (e.g., low temp). The lethal effect is only observed under restrictive conditions (e.g., high temperature), allowing for the study of the gene's function. **[3]**

14. **Explain the mechanism of F+ to F- conjugation in bacteria. What is the role of the F plasmid in this process [U] [CO2]**

Mechanism of F+ to F- Conjugation: An F+ (donor) cell extends a sex pilus and attaches to an F- (recipient) cell, forming a conjugation canal and enabling plasmid mobilisation **[2].** Rolling circle replication occurs, and the recipient cell becomes an F+ cell, now capable of acting as a donor **[1].** Role of F Plasmid- Carries tra genes to form the sex pilus and initiate transfer. It is the DNA molecule transferred, making the recipient a new donor. **[2]**

15. **Explain the mechanism by which gamma rays cause DNA mutations. [U] [CO3]**

Gamma rays are a form of ionising radiation.**[1].** They cause double-stranded breaks in the DNA backbone.**[2].** The cell's repair mechanisms are error-prone, leading to deletions, insertions, or chromosomal rearrangements**.[2]**

16. **Explain composite transposons with an example [K]/[CO1]**

A composite transposon consists of two Insertion Sequences (IS) elements flanking a central region.**[1].** The central region contains additional genes (e.g., for antibiotic resistance).**[1].** The two IS elements can be in direct or inverted repeat orientation **[1].** The IS elements provide the transposase enzyme needed for movement **[1].** Any **Example: Tn** 10 (confers tetracycline resistance), flanked by two IS10 elements.**[1]**

17. **Illustrate the sequence of events that occur when *E.coli* is transferred from a glucose-rich medium to a lactose–only medium [U] [CO2]**

Glucose is depleted. cAMP levels rise and bind to CAP(CRP), activating it [**1].** The **lac repressor** is inactivated by binding to the inducer (lactose) **[1]**. The cAMP-CRP complex binds to the promoter region, thereby enhancing RNA polymerase binding. Transcription of the lac operon (lacZ, lacY, lacA genes) occurs. Translation produces the enzymes β-galactosidase (metabolises lactose) and lactose permease (transports lactose). **[2].** Diagram **[1]**

18. **Distinguish between substitution, insertion, and deletion mutations with one example each. [U] [CO3]**

Substitution - One base pair is replaced by another. Sickle-cell anaemia (A → T) or Rifampin resistance in *M. tuberculosis* **[2].** Insertion - One or more base pairs are added into the DNA sequence. Example: Frameshift mutation in the *E. coli* lacI gene, inactivating the repressor.**[1.5]** Deletion (1.5 Marks): One or more base pairs are removed from the DNA sequence. Any example - Deletion in a viral genome leading to an attenuated vaccine strain. **[1.5]**

Part C – Long Essay - Answer Any 1. 1X10 = 10

19. **What are the applications of plasmids? [K]/[CO1]**

Introduction to plasmids **[3]**. Seven applications include: Cloning: Serve as vectors to insert and replicate foreign DNA in a host cell. Protein Production: Used to express large quantities of recombinant proteins (e.g., insulin, vaccines). Gene Therapy: Acts as a vector to deliver therapeutic genes to human cells. Used to study gene function and regulation in molecular biology. Antibiotic Resistance Markers: Enable the selection of successfully transformed bacteria. Synthetic Biology: Act as standardised parts to build novel genetic circuits. DNA Sequencing. Gene Knockdown: Basis for constructing plasmids for RNA interference. Bioremediation. Used to create genetically modified (GM) crops with desirable traits.**[7]**

20. Explain the key components involved in the induction and repression of gene expression in prokaryotes [U] [CO2]

Structural Genes - lacZ, lacY, lacA). They are transcribed together as a single polycistronic mRNA. **[1].** Promoter - DNA sequence where RNA polymerase binds to initiate transcription. **[1].** Operator - regulatory switch - Binding of a repressor protein to the operator blocks RNA polymerase. **[1].** Repressor **-** a regulatory protein that binds to the operator to prevent transcription. Inducible System (e.g., lac operon): The repressor is active alone and binds the operator, blocking transcription. An inducer binds to the repressor, inactivating it and allowing transcription**.[2].** Inducer and Corepressor **[1.5].** Activator Protein (CAMP – CRP) **[1.5].** Overall Mechanism with a diagram **– [2]**